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Experimental and numerical investigations on the spray characteristics of liquid-gas pintle injector

Wenyuan Zhou [a,b], Xu Xu [a,*], Qingchun Yang [a,*], Ronghui Zhao [c], Yushu Jin [a,d]

[a] School of Astronautics, Beihang University, Beijng, 100191, China
[b] Shanghai Institute of Space Propulsion, Shanghai, 201112, China
[c] Beijing Power Machinery Institute, Beijng, 100074, China
[d] Hypersonic Aerodynamics Institute of CARDC, Mianyang, 621000, China

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ABSTRACT

The spray characteristics of a liquid-gas pintle injector at different total momentum ratios (*TMR*) are investigated by means of experiment and numerical simulation. Water and nitrogen are used as simulant for liquid fuel ejected out from radial gap and gas oxidizer ejected out from axial annular gap, respectively. A high-speed photography system and Malvern measuring instrument are used to study the spray pattern, spray angle, droplet diameter and spatial distribution. A coupling algorithm between Volume of Fluid and Lagrangian Particle Tracking (VOF-LPT) is applied for spray simulations implemented within the CFD software OpenFOAM. The results indicate that the spray pattern generated by the liquid-gas pintle injector is mainly influenced by gas weber number and divided into three types. The spray angle is dominated by the *TMR*, and the outer boundary of the spray field is enlarged with the increase of the *TMR*. In particular, a spray angle formula of liquid-gas pintle injector is given in: $\cos\theta = c/(c + TMR)$, and $c = 0.71$ with considering the gas jet momentum loss. The droplet diameter and spatial distribution indicate that Sauter Mean Diameter (SMD) decreases in the spray intermediate area with the increase in the aerodynamic force action. The spray structure and diameter distribution obtained by numerical calculation agree well with the experimental results, and can be used to analyze the spray characteristics.
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1. Introduction

The pintle injector is developed for applications in variable thrust rocket engine with the advantages of simple design, inherent combustion stability and deep throttling capacity [1]. The spray of pintle injector is formed by the collision of radial jets of fluid through the center of the pintle post with an annular sleeve of fluid travelling axially along the post.

The pintle injector originated from the early laboratory experimental apparatus at the Caltech Jet Propulsion Laboratory (JPL) [2]. To date, it has been applied into rocket engines over a wide range in thrust and operation with 25 different propellant combinations. The most notable achievement of pintle injector technology is the lunar module descent engine (LMDE) [3][4][5], which is used to the moon landings and return the astronauts to earth orbit. Despite of the pintle injector was initially designed for variable thrust engine, it was also applied into fixed thrust engine due to the excellent performance and low cost, such as TR-106 [6][7] and Merlin series [8]. The Merlin engine is developed by SpaceX to use on Falcon launch vehicles for sea recovery and reuse. The pintle injector was used for gel propellant tactical missile in the Future Missile Technology Integration (FMTI) program [9] because of the face shut-off (FSO) operating feature. This technology prevents injector plugging and allows extremely short firing times and pulse widths. The pintle injector is also applied in the small engines nontoxic hypergolic propellants based on hydrogen peroxide and a methanol-based fuel [10].

Despite of the pintle engine extensive hot tests and wide application [11][12][13][14][15][16][17][18][19], the fundamental studies focusing on the pintle injector are rather limited [20]. According to the optical measurements [21][22][23][24][25] and numerical investigations [26][27][28][29][30][31], the flame structure and combustion performance of engine are highly dependent on the spray structure. Most of previous studies on pintle injectors have focused on the spray angle. For liquid-liquid pintle injector, Heister (2011) [1][32] considered the spray angle increases with the total momentum ratio (*TMR*), scales as $TMR^{0.5}$ is similar to the scaling of jet penetration in a crossflow with momentum ratio. Santoro et al. (1999) [33] set up cold flow studies on the sprays of three pintle

* Corresponding authors.
E-mail addresses: xuxu_1521@126.com (X. Xu), yangqingchun@buaa.edu.cn (Q. Yang).

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Nomenclature

C_D	particle drag coefficient	
D	pintle tip diameter	mm
d_l	radial flow opening distance	mm
d_g	axial flow annular gap thickness	mm
d_p	particle diameter	mm
L_s	skip distance	mm
p	pressure	Pa
Re_p	particle Reynolds number	
SMD	Sauter Mean Diameter	
TMR	Total momentum ratio	
u	fluid velocity	m/s
$u_{p,i}$	particle velocity	m/s
We_l	liquid Weber number	
We_g	gas Weber number	
x_p	particle center	
α	volume fraction	
ρ	density	kg/m^3
ρ_l	liquid density	kg/m^3
ρ_g	gas density	kg/m^3
ρ_p	particle density	kg/m^3
μ	viscosity	
μ_l	liquid viscosity	
μ_g	gas viscosity	
σ	surface tension	
θ	spray angle, deg	
κ	surface curvature	
ψ_p	droplet sphericity	

injectors with discrete radial orifices. The measurements indicated that as the *TMR* increases, the spray angle increases at a less than linear rate. Cheng et al. (2017) [34] proposed a spray half angle formula: $\cos\theta = 1/(1 + TMR)$ based on a control volume analysis. The theoretical study was validated with numerical results and actual experiments. Cheng et al. [35] designed a pintle injector element to studied the flow characteristics of the pintle injectors with discrete radial orifices based on the theoretical analysis and experiments. It was found that the spray pattern and breakup are heavily related to the local momentum ratio (LMR) and Reynolds numbers of jet and film. Ninish et al. (2018) [36] studied the interaction between a thin annular sheet and radial jets of liquid in pintle injectors with laser shadowgraphy. The spray pattern was investigated for different *TMR* and the influence on the droplet formation and size distribution was demonstrated. Chen et al. (2019) [37] used a high-speed camera and Phase Doppler Anemometry (PDA) system to investigated the spray pattern, spray angle, droplet size and diameter-velocity distribution of an oxidizer-centered pintle injector.

For liquid-gas pintle injector, Son et al. (2015) [38] put forward the spray angle had an exponentially decreasing correlation as a *TMR* divided by the Weber number. Son et al. (2017) [39] investigated the spray angle and Sauter Mean Diameter (SMD) experimentally by changing the geometry design parameters. Lee et al. (2019) [40] performed to simulate cryogenic spray of liquid oxygen and gaseous methane by cryogenic liquid nitrogen and gaseous nitrogen. Experimental conditions were varied by changing the pintle tip angle, pintle opening distance, and simulant supplying pressure. Fang et al. (2017) [41] studied the influence of gas-liquid mass flow ratio on atomization cone angle, and measured SMD and spread parameter in different axial positions. Erkal et al. (2019) [42] investigated the cone angles and SMD of manufactured injector at different throttle levels with high-speed shadowgraph system and PDPA system. Lee et al. (2020) [43] designed a pintle injector to investigated the radial flow orifice size and the orifice area for annular flow influence on spray characteristics such as spray angle, spray uniformity, and droplet size. Nevertheless, compared to liquid-liquid pintle injector, the recent studies of the spray characteristics of liquid-gas pintle injector are not comprehensive and the spray patterns have not been reported. The breakup of the liquid propellant determines the mixing of fuel and oxidizer, and has a significant influence on the combustion efficiency and thrust performance [15][16]. Therefore, an in-depth understanding of the spray characteristics is extremely necessary for the optimum design of liquid-gas pintle injector and engine.

In previous references, fewer simulation studies for the pintle injector have been published in comparison with the experiments. Son et al. (2016) [44] conducted the numerical simulation showed that the spray angle decreased with the increase in the gas to liquid momentum ratio by the Eulerian multiphase model. Radhakrishnan et al. (2018) [45][46] researched on the influence of the injection conditions on the mixing performance numerically with the Lagrangian approach. Zhang et al. (2020) [47] studied the mechanism of mixing and atomization experimentally and numerically based on the Lagrangian method. Since the Euler-Lagrange system cannot capture the primary breakup of liquid jet and depends on the accuracy of the initial droplet distribution settings, a coupling numerical method is used in present study to simulate the primary breakup and secondary breakup processes of liquid-gas pintle injector.

In this paper, the spray characteristics of a liquid-gas pintle injector are investigated by experiment and numerical simulation. The high-speed photography system is used to record the spray pattern and spray angle. The Malvern measuring instrument is adopted to obtain the droplet size distribution. A three-dimensional VOF-LPT coupling algorithm is applied for spray simulations implemented within the open source CFD library OpenFOAM v2006 [48]. A series of numerical simulations have been conducted in parallel to compare with the results of the actual flow field.

2. Experimental setup and method

The cold flow experiments are conducted at atmospheric conditions. Fig. 1 shows the experimental setup. Apart from the supply system, measurement system, control system and the pintle injector, the high-speed photography system and Marvin measuring instrument are also equipped to record the spray field and measure the diameter of droplets simultaneously.

According to reference [28], a fuel-centered liquid-gas pintle injector is employed. The designed fuel and oxidizer are kerosene and GO_2 respectively. Kerosene flows down the center of the post and out radially through the slot, while GO_2 flows in a thin annular sheet axially along the post. The oxidizer travels downstream before colliding with the fuel and the distance is defined as the "skip distance" [1]. Filtered water is pressurized by high pressure nitrogen and fed to the radial slit as simulant for kerosene. The stepper motor drives the pintle to adjust the opening distance of the radial flow. Pure nitrogen is controlled by needle regulating valve as simulant for GO_2. Pressure sensors with an accuracy of 0.2% full scale (FS) are used to measure the pressure in liquid and gas manifold, and the full-scale output for the pressure sensors are both 10 MPa. The mass flow rate is measured by a turbine flowmeter with an accuracy of 0.5% FS. The full-scale output of the turbine flowmeter is

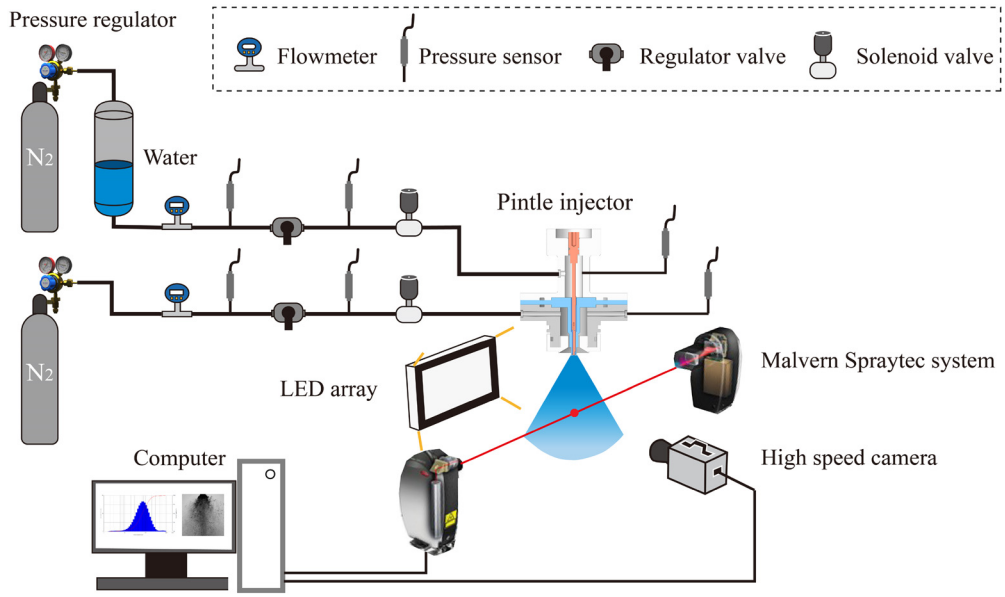


Fig. 1. Schematic of the experimental setup.

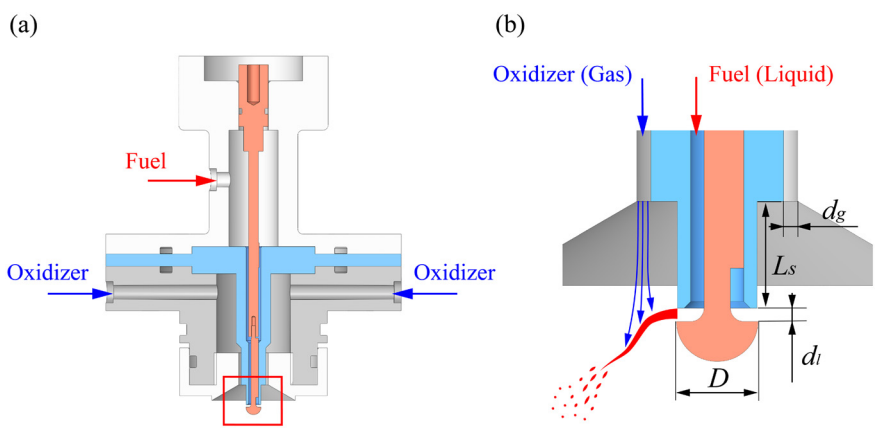


Fig. 2. Schematic diagram of the pintle injector.

Table 1
Structural parameters of the pintle injector.

Pintle tip diameter D/mm	6
Pintle opening distance d_l/mm	0 ∼ 2
Annular gap thickness d_g/mm	1
Skip distance L_s/mm	8

1000 g/s. The schematic diagram of the pintle injector is shown in Fig. 2 and the detailed parameters of the pintle injector are listed in Table 1.

2.1. Experimental conditions

Three non-dimensional characteristic parameters are defined as follows, the gas Weber number [49]:

$$We_g = \frac{\rho_g \left(v_g - v_l\right)^2 d_g}{\sigma} \tag{1}$$

the liquid Weber number:

$$We_l = \frac{\rho_l v_l^2 d_l}{\sigma} \tag{2}$$

and the total momentum ratio (*TMR*), which is defined as the ratio of the radial-to-axial stream momentum:

$$TMR = \frac{\dot{m}_r v_r}{\dot{m}_z v_z} \tag{3}$$

where $\dot{m}$ and v represent the mass flow rate and velocity respectively. The subscript "z" and "r" indicate the axial and radial direction respectively.

The effects of *TMR* and gas Weber number on spray characteristics are investigated by varying the mass flow rate of the gas under a constant mass flow rate of the liquid. Three operating conditions are set for comparison in this research as listed in Table 2, including the mass flow rate of the radial/axial flow, the total momentum ratio (*TMR*), the liquid/gas Weber number and pintle opening distance.

2.2. High-speed photography system

The high-speed photography system is adopted to visualize the instantaneous spray, which consists of an LED array, a high-speed camera and a control computer. The LED array (700 × 400 mm, maximum power: 450W) provides a stable background light and the high-speed camera (Photron Nova S12) equipped with a Nikon AF 60 mm micro lens is used to record the images. The camera exposure time is 1/500000 s, and the image resolution is 1024 × 1024 pixels. The frame frequency is 12800 fps.

Table 2
Experimental and numerical operating conditions.

Operating conditions	Radial mass flow rate/(g/s)	Axial mass flow rate/(g/s)	Total momentum ratio (*TMR*)	Liquid Weber number We_l	Gas Weber number We_g	Pintle opening distance/mm
Test 1	22.4	6.77	0.753	1864	216	0.1
Test 2	22.6	9.57	0.205	1915	978	0.1
Test 3	22.8	20.5	0.034	2038	3186	0.1

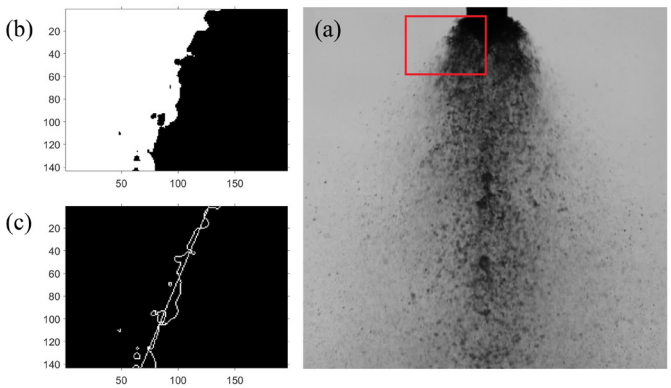


Fig. 3. Acquisition process of the spray angle.

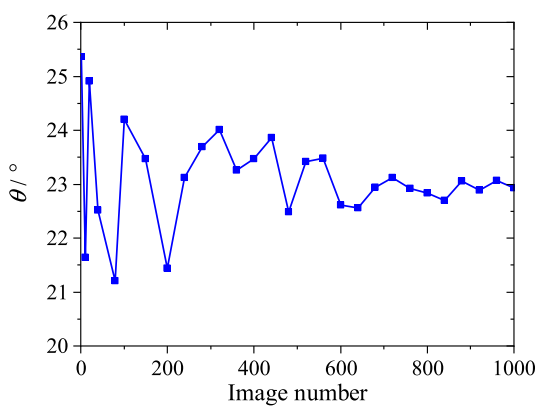


Fig. 4. Average value of the spray angles for different number of images processed.

The backlit photography technique is employed to obtain the spray angle. As the background light passes through the spray, the light intensity is attenuated by the scattering and absorption of the liquid droplets. An original spray image is shown in Fig. 3(a), and the area ranges from the radial slot to downstream is selected. Subsequently, the gray-level image is converted to binary image. The boundary between black and white is spray liquid-gas boundary, which is fitted by ordinary least squares techniques as shown in Fig. 3(c). In this study, the spray angle (θ) is defined as the angle between the spray liquid-gas boundary and the vertical axis. To reduce errors during image processing, a group of spray angle of images in certain operating condition are obtained firstly, and then an average is computed. The average value of the spray angles for different number of images processed is shown in Fig. 4. With increase of images processed, the average values tend to be stabilizing. For each operating condition, 1000 images are selected to obtain the spray angle. This image acquisition system is also applied to capture the breakup process of liquid sheet to analyze the spray pattern of the interaction between liquid jet and gas flow.

2.3. Malvern measuring instrument

The Malvern laser diffraction measurements (Spraytec) are used to measure the droplet size distribution. The Malvern droplet sizing is conducted to capture the intensity and pattern of the scattered light resulting from a laser beam passing through the region of dispersed droplets.

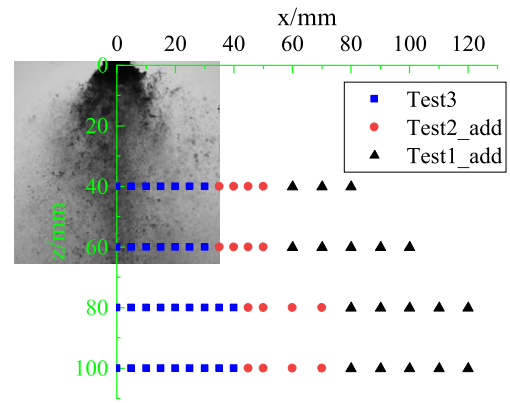


Fig. 5. Malvern measurement point setup.

The laser wavelength is 632.8 nm, the lens of the receiver is 300 mm, and the laser beam is 10 mm. The sampling period is set at 1 s, and the data acquisition rate is set 10 kHz. The spray developments under different operating conditions are visualized by the high-speed photography system and the spray field is estimated to decide the measurement grid, which is set up more location downstream, as shown in Fig. 5.

The Sauter mean diameter (SMD) is expressed in Eq. (4) as a radio of volume to surface, and is a representative diameter that reflects evaporation rate and combustion reaction [43]:

$$\mathrm{SMD} = \frac{\sum_{i=1}^{n} d_i^3}{\sum_{i=1}^{n} d_i^2} \tag{4}$$

where d_i represents the diameter for the ith droplet being counted.

3. Numerical approach

3.1. Volume of fluid method

The two-phase flow is modeled with the Volume of Fluid (VOF) method by Hirt and Nichols (1981) [50]. Both the gas and liquid phases are considered incompressible and immiscible, and the governing equations for mass and momentum are expressed as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \boldsymbol{u} = 0 \tag{5}$$

$$\frac{\partial \rho \boldsymbol{u}}{\partial t} + \nabla \cdot \rho \boldsymbol{u}\boldsymbol{u} = -\nabla p + \rho \boldsymbol{g} + \nabla \cdot \left[\mu \left(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T\right)\right] + \sigma \kappa \nabla \alpha \tag{6}$$

where $\boldsymbol{u}$, ρ, p, μ, σ, κ and α are the fluid velocity, density, pressure, viscosity, surface tension, surface curvature and the volume fraction, respectively. The terms on the right hand are the pressure gradient term, the gravity, the viscous stresses, and the surface tension, respectively. The surface tension force is modeled with the Continuum Surface Force (CSF) approach of Brackbill et al. (1992) [51] and the curvature of the interface is calculated from the volume fraction α as:

$$\kappa = -\nabla \cdot \left(\frac{\nabla \alpha}{|\nabla \alpha|} \right) \tag{7}$$

The VOF method adds one governing equations for the transport of the volume fraction α, the advection equation:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot \alpha \boldsymbol{u} = 0 \tag{8}$$

The step function α marks the volume fraction of the tracked phase in a control volume, so that $\alpha = 1$ corresponds to a control volume entirely occupied by liquid phase and $\alpha = 0$ corresponds to a control volume not containing any liquid phase, only gas phase. The interface between the phases is found in cells where $0 < \alpha < 1$.

In order to solve the advection equation for the volume fraction, a geometric VOF scheme, isoAdvector (Roenby et al., 2016) [52] is used based on the octree spatial discretization. Interface reconstruction algorithm is performed using a piecewise-linear interface reconstruction method (PLIC) coupled with the reconstructed distance function (RDF) in each cell [53]. Once interface reconstruction has been performed, direction split geometrical fluxes can be computed easily on regular Cartesian grids. This geometric method can greatly preserve sharp interface and keep quality conservation by using the concept of isosurfaces to calculate more accurate face fluxes for the cells containing the interface.

The function α makes it possible to use only one set of equations in the entire flow domain for describing the local properties, instead of one set of equations for each phase. Fluid properties are calculated using α as a weight. The density and dynamic viscosity of the two-phase mixture can be obtained by:

$$\rho = \alpha \rho_l + (1 - \alpha) \rho_g \tag{9}$$

$$\mu = \alpha \mu_l + (1 - \alpha) \mu_g \tag{10}$$

where both the liquid and gas densities and dynamic viscosities are constant.

3.2. Lagrangian particle tracking method

The motion of droplets in gas is described with the Lagrangian Particle Tracking (LPT) approach. The positions and velocities of the particle p are updated by applying Newton's second law:

$$\frac{dx_{p,i}}{dt} = u_{p,i} \tag{11}$$

$$m_p \frac{du_{p,i}}{dt} = \pi \mu d_p \frac{C_D Re_p}{8} \left(u_i - u_{p,i} \right) + \frac{\pi d_p^3}{6} \left(\rho_p - \rho \right) g_i \tag{12}$$

where x_p, d_p, $u_{p,i}$, ρ_p, C_D and Re_p are the particle center, diameter, velocity, density, drag coefficient and Reynolds number, respectively. The last term is the combination of buoyancy and gravity forces and the term before that is the drag force. The drag coefficient is calculated based on the Schiller-Naumann drag model [54]:

$$C_D = \begin{cases} \frac{24}{Re_p} \left(1 + 0.15 Re_p^{0.687} \right) \; if \; Re_p \leq 1000 \\ 0.44 \; if \; Re_p > 1000 \end{cases} \tag{13}$$

with particle Reynolds number Re_p is given by

$$Re_p = \frac{\rho_p d_p \left| u_i - u_{p,i} \right|}{\mu_p} \tag{14}$$

The droplets breakup due to aerodynamic force is modelled using the Reitz-Diwakar secondary breakup model [55] and the collisions and coalescence between droplets is computed with the collision algorithm from Nordin (2001) [56].

3.3. VOF-LPT coupling algorithm

In the VOF method, all liquid structures are resolved by the computational grid. The liquid jet in gas flow breaks up into a large number of droplets that are orders of magnitude smaller than its diameter, leads to high computational costs. In the present study, we performed multiscale simulations of atomization process using the VOF method coupled with a two-way coupling LPT model (Heinrich et al., 2020) [57] to simulate the small droplets.

For VOF to LPT transformations contiguous lumps of the liquid are identified by a three dimensional four-connectivity spatial Connected Component Labelling (CCL) algorithm. The diameter of droplets is computed as follows:

$$V_p = \sum_i \alpha_i V_i \tag{15}$$

$$d_p = \sqrt[3]{\frac{6 V_p}{\pi}} \tag{16}$$

where index i denotes all cells belonging to a droplet. And the droplet sphericity is calculated as follows:

$$\psi_p = \frac{2 r_{\max}}{d_p} \tag{17}$$

with the maximum distance from center of mass to the droplet free surface $r_{\max}$ and the diameter of a perfect sphere with the same mass as the given droplet d_p. If the diameter (<500 μm) and sphericity (<2) of droplets meet the transformation criteria, the VOF droplet is removed (volume fraction set to zero) and replaced by a Lagrangian particle located at the centroid of the droplet. Corresponding properties of the droplet is assigned to the representative particle.

3.4. Adaptive mesh refinement

In order to resolve the multiscale phenomenon of liquid jet breakup, the adaptive mesh refinement (AMR) procedure is used based on the automatic resolution control (ARC) algorithm (Jasak, 2000) [58]. The applied AMR on Cartesian mesh keeps the entire interface refined at the finest refinement level at every time step. One of the advantages of the AMR is that mesh refinement and coarsening could decrease the computation cost significantly with maintaining the same accuracy.

In the present simulations, the refinement is based on the value of the volume fraction and activated in the cells which contain the interface. The cell sizes, refinement level and details of the AMR grid are further discussed in Section 3.5.

3.5. Simulation settings

The simulation domain consists of the pintle injector interior and a cuboid external environment with the size is $[-60\delta, 60\delta] \times [-120\delta, 0] \times [-60\delta, 60\delta]$ in the x, y and z directions, where δ represents the annular gap thickness of gas inlet. The boundary consists of four regions: the liquid inlet, the gas inlet, the injector walls and the outlet condition for the environment as depicted in Fig. 6. At the liquid inlet, a Dirichlet condition of $\alpha = 1$ is imposed for the volume fraction, while a Dirichlet condition is used for the velocity (1.26 m/s), which is surface-normal to go inside the domain. At the gas inlet, the volume fraction is set as $\alpha = 0$, and the velocity is 320 m/s is surface-normal to go inside the domain. At the injector walls, Neumann conditions are applied for both p_{rgh} and α, while a no-slip Dirichlet condition is used for the velocity. At the outlet, a mixed type boundary condition is used. Dirichlet conditions are used for both the volume fraction ($\alpha = 0$) and the

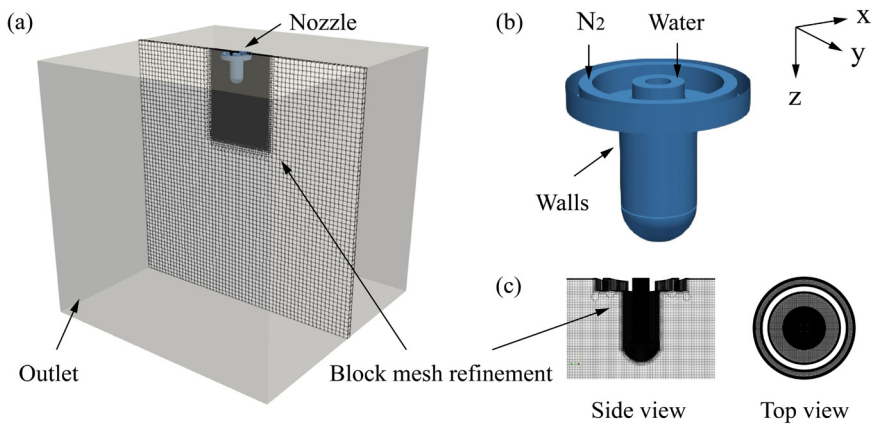


Fig. 6. (a) The simulation domain consists of the nozzle and a cube outside environment; (b) The geometry and boundary of the nozzle; (c) The BMR of the nozzle walls and the inlet channel.

Table 3
Minimum mesh size under three different resolutions.

	Grid-C	Grid-A	Grid-F
Refinement levels	5	6	7
Minimum cell size	$\delta/16$	$\delta/32$	$\delta/64$

velocity when the flow is into the domain. Neumann conditions are used for α and velocity when the flow is out of the domain.

An unstructured grid consisting of a Cartesian background grid with a block mesh refinement (BMR) is employed. The background grid has 60 × 60 × 60 cells. The region inside the injector is refined in 6 refinement levels with a cell size of $\delta/32$ that the width of the first cell is $y^+ \approx 1$. The initial grid has approximately 6.3 million cells and does not include the AMR mesh. Time-step adaptation is used in order to keep the maximum Courant number at the value 0.3 in the simulations with AMR.

Three different resolutions of the grid are used to simulate the case 1 in Table 3. The volume fraction isosurface image in Fig. 7 compares the liquid sheet structures captured in Grid-A, Grid-C and Grid-F. Grid-A is very close to Grid-F and demonstrates more detail features than Grid-C. The continuous liquid sheet with surface waves, liquid ligaments and droplets of the primary breakup process are captured in Grid-A and Grid-F. However, the liquid sheet disintegrates into ligaments directly in Grid-C due to the low grid resolution. Fig. 8 indicates the liquid mass of simulation domain of theoretical value and simulation results in different grids. Grid-A and Grid-F are very close to theoretical value with the relative error of 1.4%. The liquid mass of Grid-C is smaller significantly, and the relative error between theoretical value and simulation result in Grid-C is 4.7%. Therefore, the current grid (Grid-A) is employed for all cases considering the balance between the demand of space resolution and utilization efficiency of computation resources.

The numerical simulation of liquid jet breakup is solved using interIsoFoam solver (Roenby et al., 2016) [52] in OpenFOAM®. First, the advection equation for α is solved, after which the material properties are updated and used in the momentum equation. The coupling between the velocity and pressure terms of the momentum equation is handled with the pressure-based implicit splitting of operators (PISO) algorithm (Issa, 1986) [59]. A crucial algorithmic component for the simulations of interaction of liquid sheet and gas flow is the turbulence model. For this, the one-equation eddy viscosity model (Yoshizawa, 1986) [60] coupled with the VOF method has been applied for two-phase turbulence modeling.

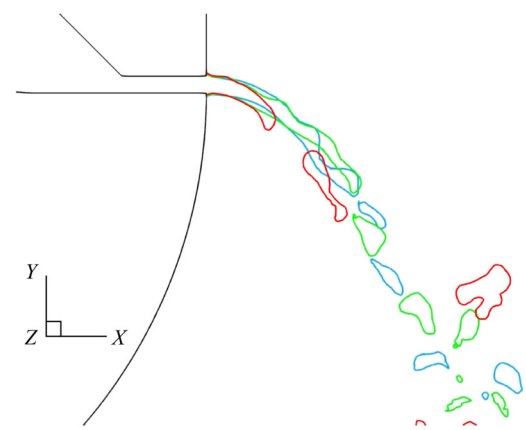


Fig. 7. Simulated liquid sheet shapes on Grid-C (red line), Grid-A (green line), and Grid-F (blue line) for Case 1.

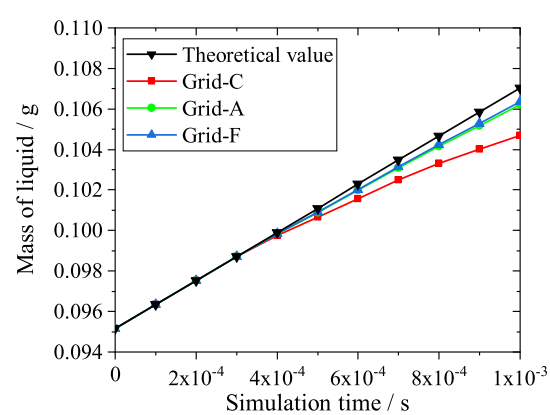


Fig. 8. Comparison of liquid mass between theoretical value and simulation results under three different resolutions.

4. Results and discussion

4.1. Spray pattern

4.1.1. Description for the spray pattern

The spray patterns for different experimental conditions are displayed in Fig. 9. It is noted that the spray generated by liquid-gas pintle injector is shaped like a solid cone with an expansion outer boundary, which differs from that of traditional injectors, such as hollow cone spray (swirl injector) [61][62][63][64][65][66]

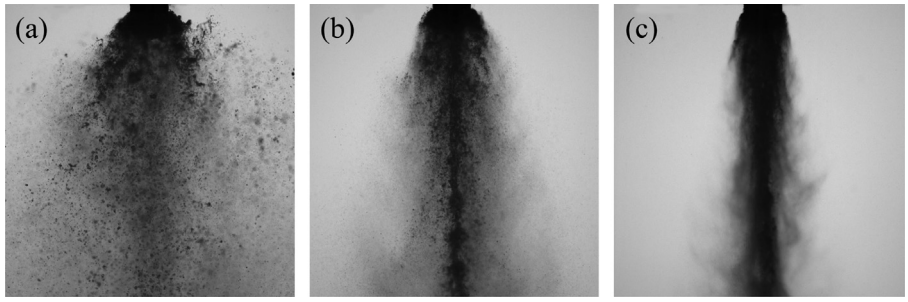


Fig. 9. Spray patterns under different operating conditions; (a) Test 1, (b) Test 2, (c) Test 3.

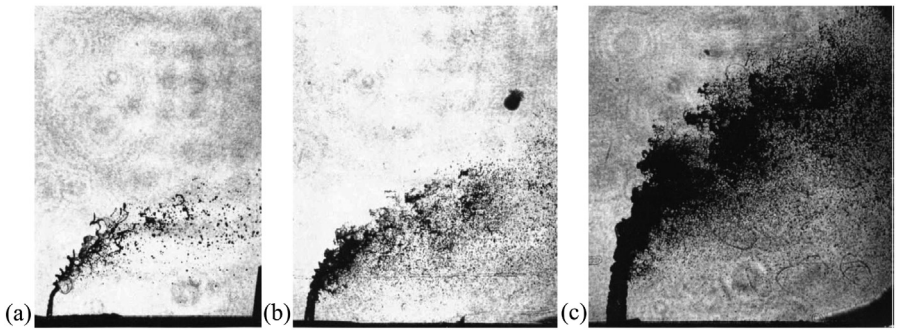


Fig. 10. Spray structures of LJICF [69].

[67], leaf-like sheet (impinging injector) [68] and cloak-like spray (liquid-liquid pintle injector element) [35].

The spray of a liquid jet in a crossflow (LJICF) [69][70] (Fig. 10) is similar to that of the liquid-gas pintle injector because both sprays are generated by the interaction of liquid jet and gas crossflow. And the spray edge of the LJICF is less distinguished than that of the liquid-gas pintle injector. However, the liquid jet of the pintle injector is ejected from a continuous annular gap, whereas that of the LJICF flows through a circular orifice [71]. And the spray of the pintle injector is axisymmetric and in the shape of a conical jet stream, whereas that of LJICF changes from a round circular to an elongated sheet-like shape. Moreover, the spray of the LJICF develops near the wall, whereas that of the pintle injector flows along the surface of the tip and gather in the center of the spray. Furthermore, the spray of the pintle injector undergoes a gathering process with the axis flowrate increase and the center of the spray becomes denser.

The spray structure is the main characteristic for spray pattern, which can be divided into three types:

Type-I: Disperse large droplet spray (Fig. 9a): For intermediate axis flow We_g (216), the conical liquid sheet keeps continuous within a short distance from the outlet. In the current study $We_g < 10^2$ are considered small; while, $We_g > 10^3$ are considered large. Some waves appear on the liquid sheet surface by aerodynamic force, and increase dramatically in amplitude as they progress downstream, which leading to the sheet is disintegrated forming liquid ligaments [72]. The ligaments are then broken into large droplets (>100 μm), as shown in Fig. 9a.

Type-II: Solid cone spray (Fig. 9b): When axis flow We_g increases to 978, the large clumps of ligaments are separated periodically from the sheet. These ligaments decay downstream into smaller ones and droplets, whereas distinct sheet-like structures in continuous sheet shrinks and barely identified. The large droplets in the center of the spray can be seen obviously and the droplets near the spray boundary tend to be smaller.

Type-III: Dense droplet spray (Fig. 9c): While at slightly higher axis flow We_g (3186), the spray structure and droplet characteristics are changed obviously. The sheet breaks up directly into very fine droplets without fragmenting into ligaments at the spray boundary. These droplets are preferably shed off in a periodic manner as waves from the sheet edge. The spray is confined to a limited space and becomes denser visually, barely includes any large droplets.

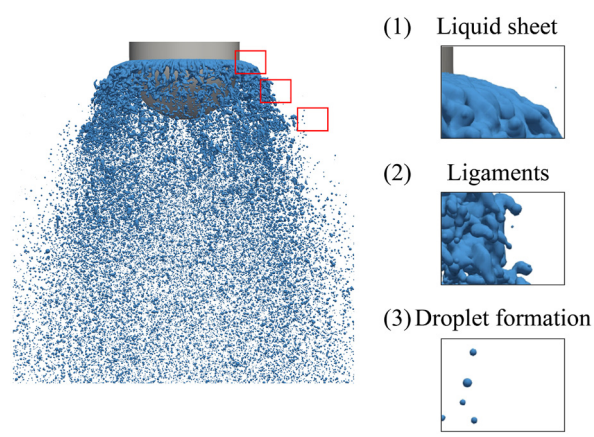


Fig. 11. Numerical spray evolution image (Test 1. operating condition).

For the Type-I spray pattern, the typical atomization process (Test 1.) is shown in Fig. 11. As can be seen, the radial liquid sheet collided with the axial gas flow, surface waves are generated by aerodynamic force, leading to the disintegration of the liquid sheet. Furthermore, the liquid ligaments fell out of the end of the liquid sheet and then split into smaller droplets.

Fig. 12 shows the liquid sheet breakup process simulation with VOF-LPT coupling. The liquid sheet and ligaments are modeled by VOF method and the droplets are identified by the coupling method and modeled by the LPT method. The conversion of small VOF droplets to Lagrangian particles reduces the computational cost significantly.

The adaptive mesh refinement based on the value of the volume fraction α is shown in Fig. 13. The interface resolution is set to be 6 level between liquid phase and gas phase. Since the flow

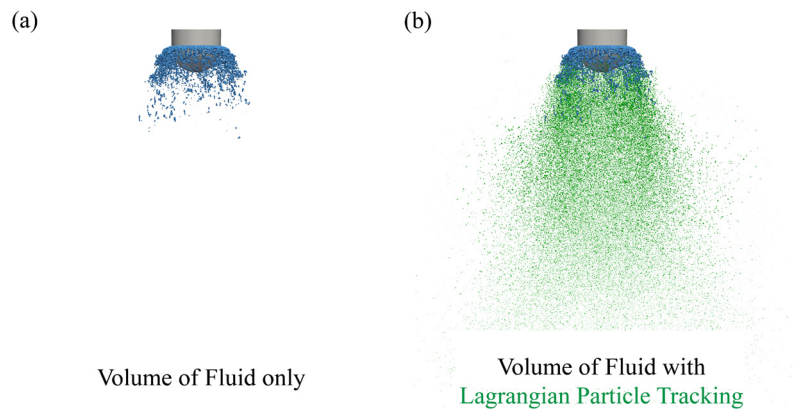


Fig. 12. Snapshot of the liquid sheet and ligaments are modeled by the VOF method (left, isosurface $\alpha = 0.5$, colored in blue) and the droplets are identified by the coupling method and modeled by the LPT method (right, color in green, scaled by the particle diameter).

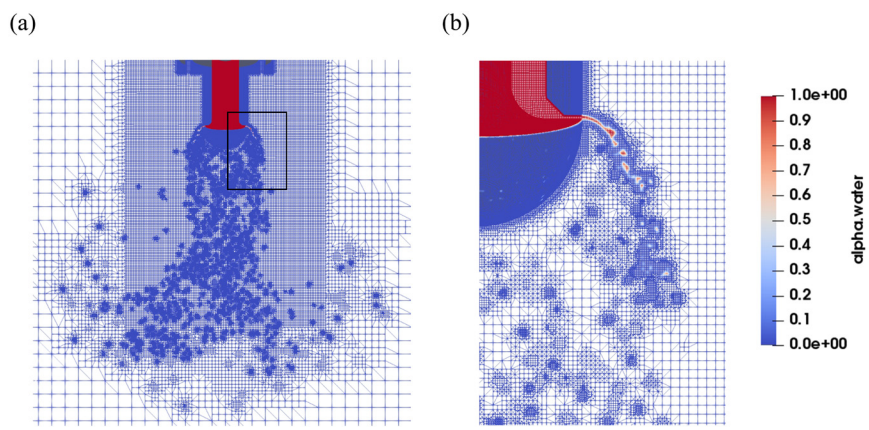


Fig. 13. AMR based on the value of the volume fraction α.

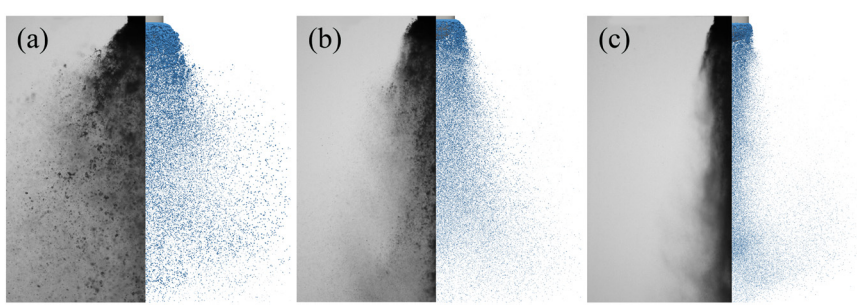


Fig. 14. Flow field of experimental (left) and numerical (right, isosurface $\alpha = 0.5$, colored in blue, scaled by the particle diameter) results; (a) case 1, (b) case 2, (c) case 3.

field is only refined locally with AMR and BMR, both computational memory and time are substantially reduced as compared to uniform meshes at the same level of resolution. For Test 1. numerical simulation, VOF-LPT coupling method with the AMR of maximum refinement level of 6, only 14.8 million cells are used, while a total of 5.6 billion cells are required for the same resolution with a uniform mesh. Less than 1% of grid points are employed in the present study, as compared to a uniform mesh.

As can be seen in Fig. 14(a) to (c), the spray structures of numerical model (right) are good accordance with experimental results (left) under different operating conditions. With the VOF-LPT coupling method and AMR procedure, the calculated shapes of the liquid sheet and ligaments agree well with experimental measurements, which are related to the spray angle and further discussed in Sec. 4.2. The spatial distribution of droplets for simulation is closed to the experiment results obtained by backlit photography, and the SMD comparison between experiment and numerical simulation is analyzed in Section 4.3.

4.1.2. *Influence of the We on the spray pattern*

The influence of the We (We_g, We_l) on the spray pattern is investigated by varying the mass flow rate of the gas, mass flow rate of the liquid and pintle opening distance, as depicted in Fig. 15. With the gas flow We_g increasing from 153 (Fig. 15c) to 4944 (Fig. 15g), the spray patterns change from the disperse large droplet spray (Type I) to the solid cone spray (Type II), and then to the dense droplet spray (Type III). The We_g is defined in Equation (1), and represents the flow parameter of the relative importance of gas inertia compared to the liquid surface tension, which is usually used for analysis the interaction between gas jet and liquid jet, such as LJICF, liquid-gas swirl coaxial injector, liquid-gas pintle injector and so on. With the enhancement of the gas jet action, the liquid sheet and ligament are more easily disconnection, the

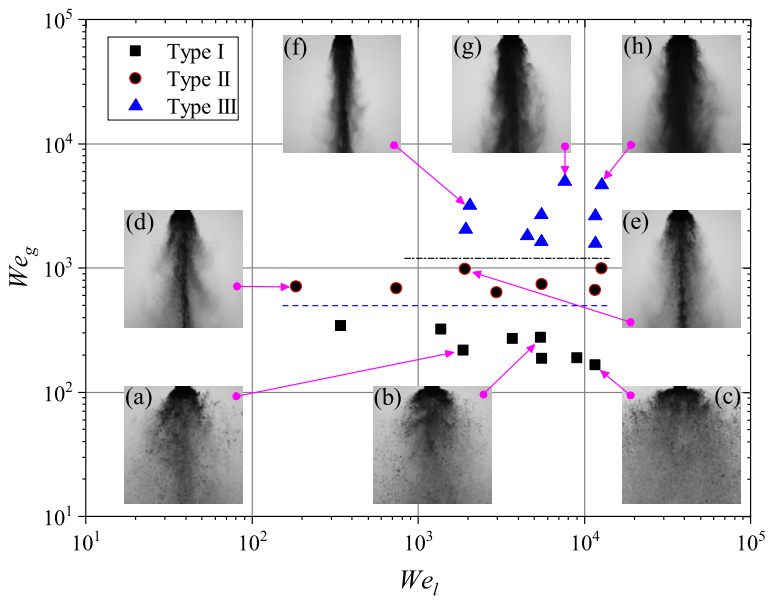


Fig. 15. Spray patterns under different operating conditions.

droplets tend to be denser, smaller and faster, and the spray tends to be confined to a smaller limited area. The conversion of spray pattern with We_g increasing indicates that the aerodynamic force is the primary factor determining spray pattern.

It is interesting that the liquid jet We_l also has influence on spray form for the same spray pattern. Compares the Fig. 15a-c in Type I, the spray expands more under the higher We_l, leading to the expansion of the solid cone downstream. For Fig. 15f-h in Type III, the sprays are always confined to a cylinder field, and the radial scale of the cylinder field also expands with the increases of the We_l. Consequently, the spray form is affected by both We_g and We_l, and the We_g is primary factor.

4.2. Spray angle

According to the experimental results and different models, the spray angle is closely related to *TMR*. Fig. 16 compares the spray angle obtained from experimental results and different models. Heister (2011) [32] considered that the spray angle scales as $TMR^{0.5}$, which is derived from liquid jet in crossflow. The curve of this model is driven by equation:

$$\theta = \frac{180}{\pi} TMR^{0.5} \tag{18}$$

As can be seen, experimental results are larger than Heister's model when *TMR* is less than 1.2, while experimental results are less than Heister's model when *TMR* is larger than 1.2. This model may not be suitable for liquid-gas pintle injector.

Boettcher et al. [73] investigated the gas-gas pintle injector and put forward a model as follows:

$$\cos\theta = \frac{1}{\sqrt{1 + TMR^2}} \tag{19}$$

This model satisfies global momentum conservation, which is used for impingement nozzle design in liquid rocket engine. It can be seen that experimental results are always larger than this model, and the differences between experimental results and this model are larger than 20° under a wide range of *TMR* conditions.

Cheng et al. (2017) [34] studied a liquid-liquid pintle injector and proposed a spray angle formula:

$$\cos\theta = \frac{1}{1 + TMR} \tag{20}$$

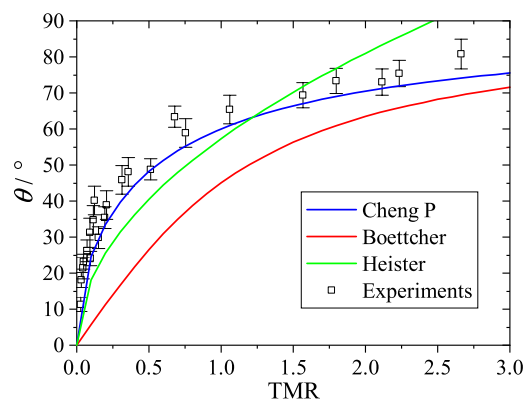


Fig. 16. Spray angle of experimental results and different models.

This equation is established to predict the spray angle according to the theoretical analysis, and fits the experimental data for liquid-liquid pintle injector reasonably well. Fig. 16 shows the experimental results of liquid-gas pintle injector are slightly larger than this model, and the trend is consistent. The difference between the liquid-gas pintle injector and liquid-liquid pintle injector results from the effects of gas jet momentum loss on the actual *TMR*.

It is noted that the friction in free surface of the liquid jet control volume is ignored in liquid-liquid pintle injector [34], due to the high liquid jet to ambient gas density ratio and low relative velocity (10 ∼ 50 m/s) [32]. For liquid-gas pintle injector, the density of gas jet is close to that of ambient gas, while the relative velocity between gas jet and ambient gas can achieve a large value (300 m/s). Therefore, the shear stress of the ambient gas to the gas jet and the momentum loss of gas jet due to the friction should be considered in present model (Fig. 17).

As mentioned previously, the effective coefficient c of gas jet momentum is define as:

$$c = \frac{I_{g,eff}}{\int_{A_{1in}} \rho_1 v_{1in}^2 dA} \tag{21}$$

where the $I_{g,eff}$ is the actual momentum of gas jet acting on the surface of a liquid jet. If the flow is assumed as inviscid, there are no boundary layer and shear layer, and then the velocities are

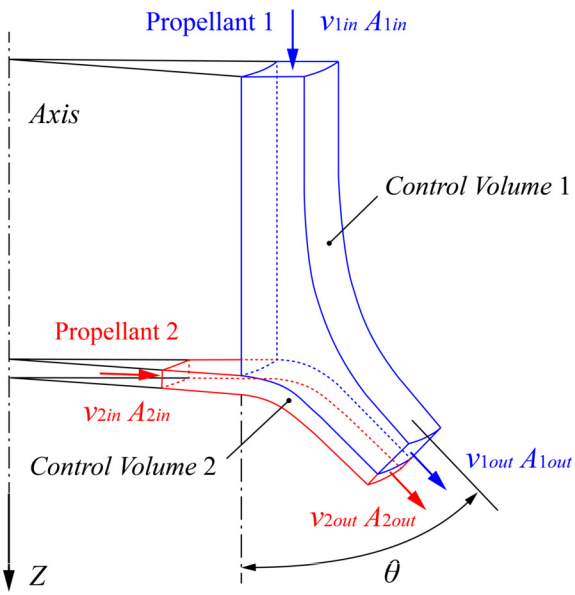


Fig. 17. Sketch of control volumes.

uniform in both inlets and outlets for two control volumes. Then Eq. (21) becomes:

$$c = \frac{I_{g,eff}}{\rho_1 A_{1in} v_{1in}^2} \tag{22}$$

Considering the continuity equation:

$$\dot{m}_1 = \rho_1 v_{1in} A_{1in} \tag{23}$$

and substitute Eq. (23) into Eq. (22) leads to:

$$c = I_{g,eff} \cdot \frac{\rho_1 A_{1in}}{\dot{m}_1^2} \tag{24}$$

Note that the *TMR* is defined as:

$$TMR = \frac{\dot{m}_2 v_{2in}}{\dot{m}_1 v_{1in}} = \frac{\dot{m}_2^2}{\dot{m}_1^2} \cdot \frac{\rho_1 A_{1in}}{\rho_2 A_{2in}} \tag{25}$$

The actual total momentum ratio *TMR*′ can be expressed as:

$$TMR' = \frac{\dot{m}_2 v_{2in}}{I_{g,eff}} = \frac{1}{c} \cdot TMR \tag{26}$$

By replacing *TMR* with *TMR*′, the prediction formula of the spray angle of liquid-gas pintle injector is given in Eq. (27), which agrees well with the experimental results, as shown in Fig. 18.

$$\cos\theta = \frac{c}{c + TMR} \tag{27}$$

where the c is defined in Eq. (21). The value of c is related to the structure of the pintle injector and relative velocities between gas jet and ambient gas. By neglecting the influence of structural parameters and different relative velocities, the effective coefficient c is assumed to a constant in present model for simplicity. The c is approximately equal to 0.71 by the least squares method based on experimental results.

Considering that the skip distance of the liquid jet travels radially before colliding with the gas jet, the effect of pintle wall could be negligible in liquid-gas pintle injector [27]. For gas jet impinging on liquid sheet, the difference between A_{1in} and A_{1out}, and the difference between A_{2in} and A_{2out} are ignored, which is the same as Cheng's model [34].

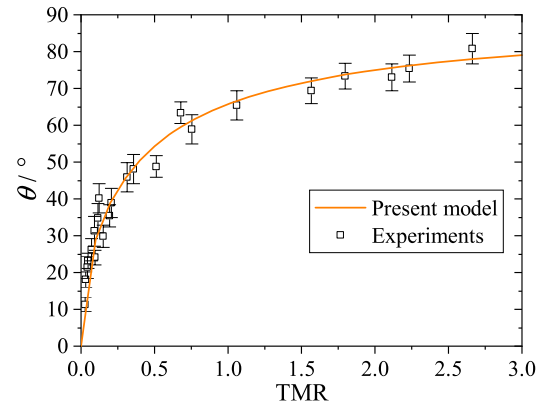


Fig. 18. Spray angle of experimental results and present model.

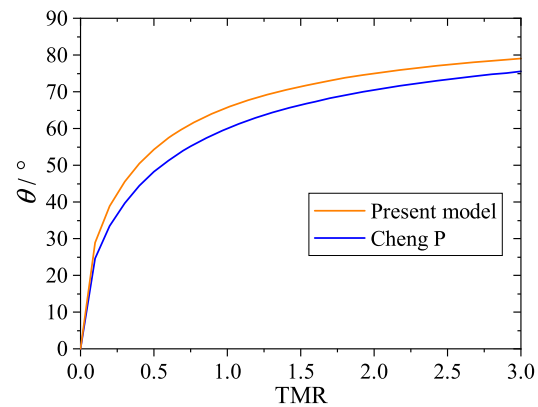


Fig. 19. Comparison of Cheng's model and present model.

Compares the spray angle obtained from Cheng's model [34] and present model. As can be seen, the curve of present model is approximate to that of Cheng's model when *TMR* is less than 0.1. The difference between Cheng's model and present model is less than 7° in the *TMR* range of 0.1-3, which indicates that the effect of gas jet momentum loss is not negligible in the liquid-gas pintle injector (Fig. 19).

4.3. Sauter means diameter

The distributions of SMD obtained by experiment under different operating conditions are analyzed in Fig. 20. The variation trends along the x direction change at different axial distances. In the spray field of z=40 mm–80 mm, the SMD decreases firstly and then increases along the x direction, indicating as a "V" shape, which is consistent with the conclusion in literature [37]. It is noted that the SMD of Test 1 (symbol-line, color in green) reaches the minimum value at a further location compared with other conditions at distance 40 mm and 60 mm away from the nozzle exit (z=40 mm–60 mm).

The spray SMD at different axial distances of Test 1 is larger than that of Test 2, while Test 2 is larger than Test 3. The reason is that the distribution of SMD is mainly influenced by aerodynamic force. With the enhancement of the air flow action, the droplet is more easily fell out of the liquid sheet and breakup into smaller ones. As can be seen from Fig. 20 and Fig. 21, the position with smaller SMD is consistent with the streamlined of gas jet. The droplets SMD just below the nozzle and near the spray boundary are larger than that in the spray intermediate area, where the air flows through. However, owing that the gas flow entrainment and the droplets lateral movement, the variation trend changes from

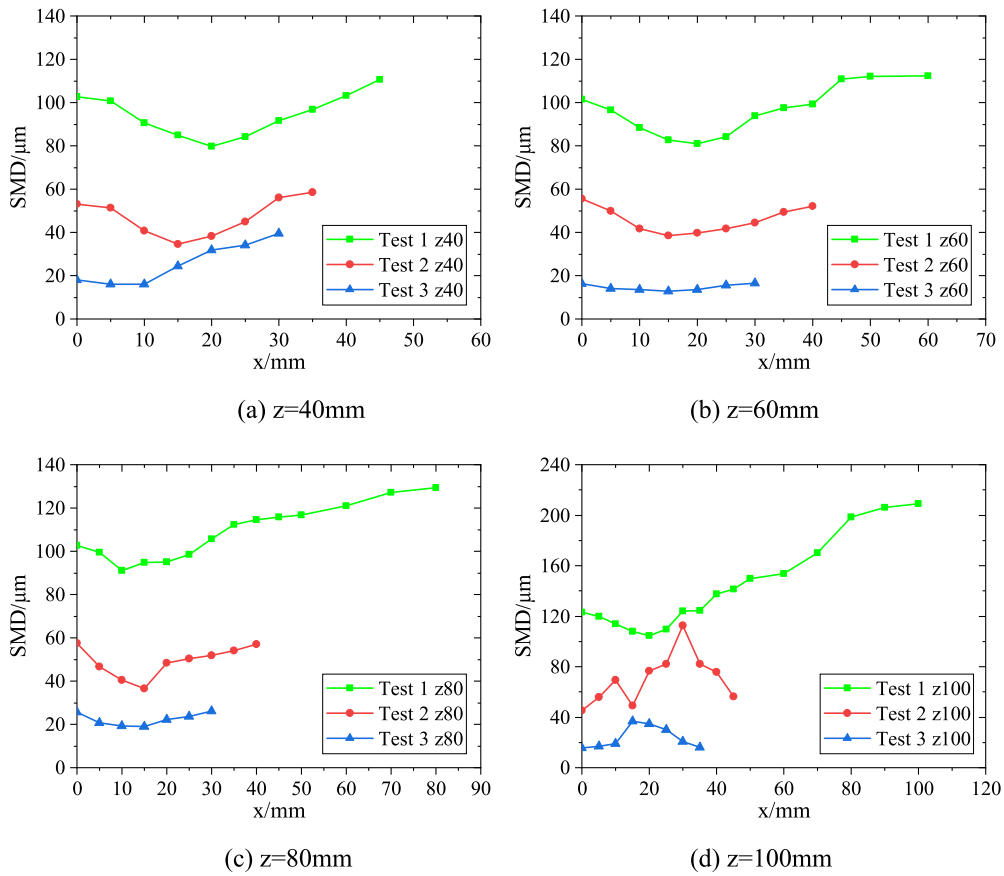


Fig. 20. Variation of SMD along x/z direction under the different operating conditions.

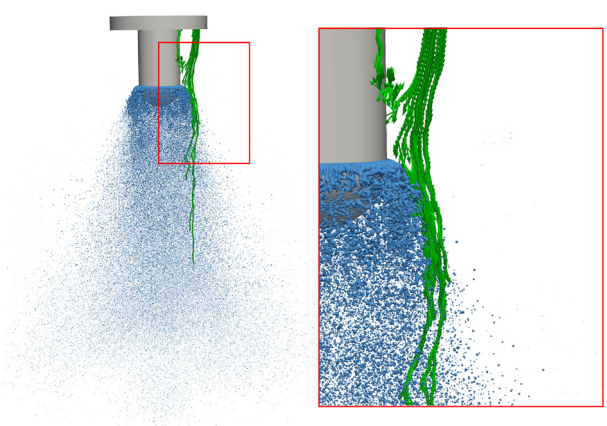


Fig. 21. Streamlines of gas-phase in the central plane (Test 2).

"V" shape into "M" shape (Test 2) and inverted "V" shape (Test 3) at far axial distance (z=80 mm).

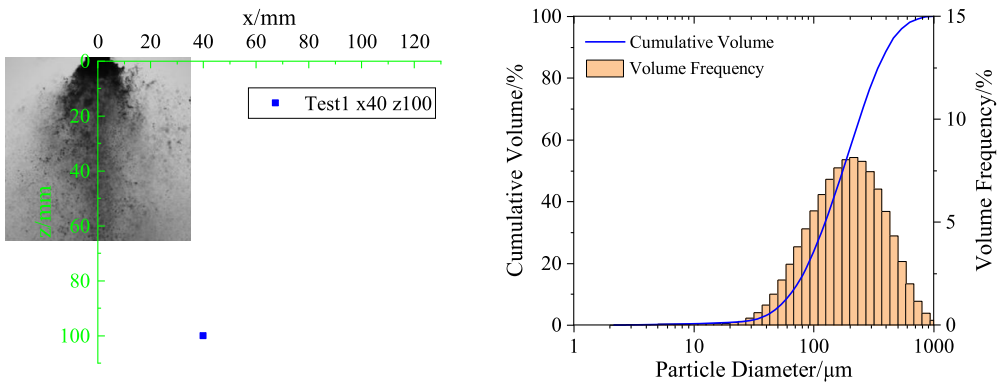
The spray field is solved by numerical method, and the streamlined of gas-phase is shown in Fig. 21, which represents the area of gas jet action. As can be seen, the gas jet is ejected out from the axial nozzle and acts on the surface of the responding sheet. Subsequently, the gas jet deflects and disperses by the shielding of liquid sheet and droplets. Consistent with the previous analysis, the gas jet mainly acts in the spray intermediate area.

Experimental measurements of droplet diameter distribution at spray downstream under different operating conditions are shown in Fig. 22. In order to summarize the key characteristics of spray distribution, the experimental results are analyzed in the position where the breakup process is fully developed. Selection of measurement points is based on the spray structure. It can be seen that the droplet diameter distribution is different and can be divided into three types. For Test 1 (Fig. 22a), the diameter of droplets ranges from 5 μm to 1000 μm, indicating as a unimodal-normal distribution. The spray of Test 1 includes both large and small droplets, leading to a large SMD, which is consistent with the disperse large droplet spray pattern.

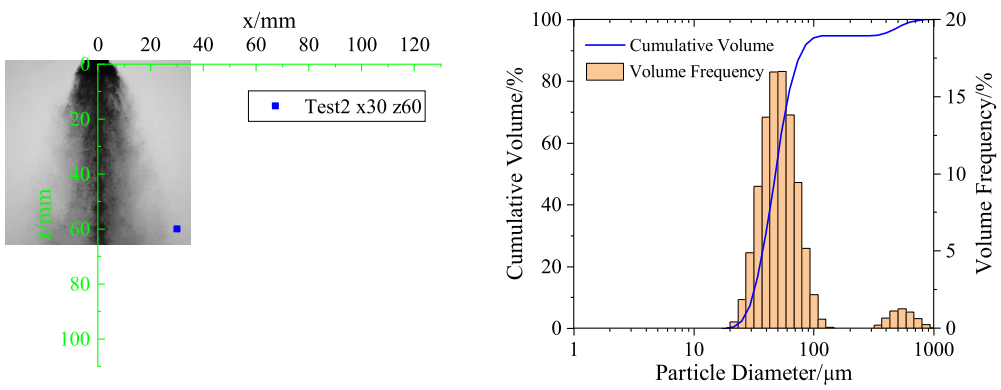
As the gas flow action increases (Fig. 22b), there are fewer large droplets and more small droplets. The droplet diameter indicating as a bimodal-normal distribution. Although the number of large droplets decreases significantly, they are always observed in repeated tests. While at slightly higher aerodynamic force action (Fig. 22c), the diameter of droplets ranges from 5 μm to 350 μm, and barely includes any large droplets. The droplet diameter returned to unimodal-normal distribution.

In order to evaluate the SMD of spray cloud from numerical results, a series of postprocessing cubes is set up at the experimental measurement point. These cubes have a side length 2 mm. Each Lagrangian particle crossing or inside the cube will get counted and used for calculating SMD.

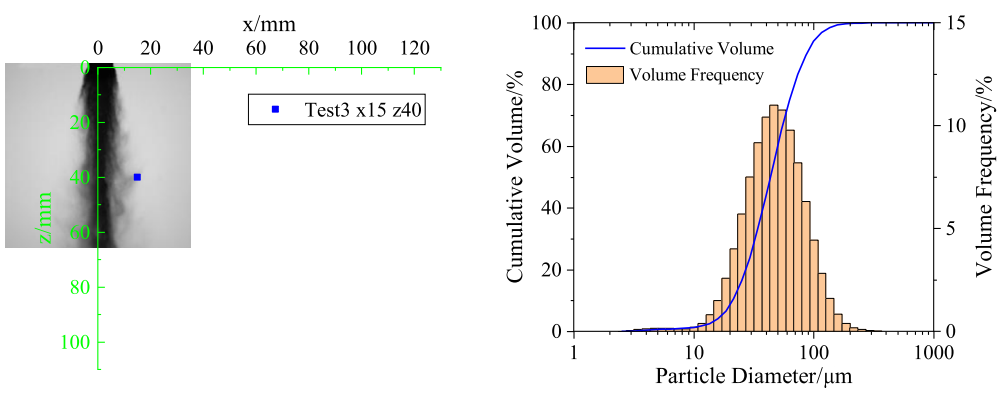
The comparison of SMD between experiment and numerical model is shown in Fig. 23. under different operating conditions. As can be seen, the SMD obtained by coupling algorithm reveals a good accuracy with the experimental measurements. This indicates that the transition from VOF droplets to Lagrangian particles modeled the distribution of SMD accurately. The high fidelity VOF method directly captures the dynamic liquid-gas interface and its topological changes of primary breakup process with proper grid resolution (Sec. 3.5). The LPT method with the proper transformation criteria (Sec. 3.3) keep quality conservation in tracking the discrete droplets of secondary breakup.



(a) Test 1. x=40mm, z=100mm



(b) Test 2. x=30mm, z=60mm



(c) Test 3. x=15mm, z=40mm

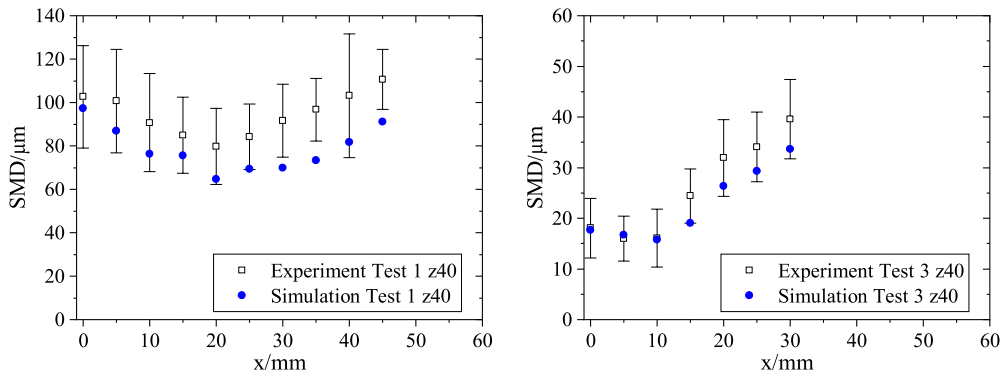
Fig. 22. Droplet diameter distribution at a measuring point of different operating conditions.

5. Conclusion

The spray pattern, spray angle, and Sauter Mean Diameter (SMD) distribution of a liquid-gas pintle injector have been investigated by experiment and numerical method, respectively. The spatial distribution and diameter distribution of spray are analyzed. The conclusions are as follows:

(1) The spray pattern is mainly influenced by gas Weber number (We_g) and divided into three types. With the increase of We_g, the spray patterns change from the disperse large droplet spray (Type I) to the solid cone spray (Type II), then to the dense droplet spray (Type III). The liquid Weber number (We_l) also have influence on spray form for the same spray pattern.

(2) While increasing the *TMR*, the spray field boundary is enlarged, and spray angle also increases accordingly. In particular, the spray angle of liquid-gas pintle injector is given in: $\cos\theta = c/(c + TMR)$ including the gas flow momentum loss, and $c = 0.71$ under the present operation conditions.



(a) Test 1. z=40mm (b) Test 3. z=40mm

Fig. 23. Comparison of SMD between experiment and simulation at different operating conditions.

(3) The SMD distribution exhibits different trends along the x direction at different axial distances. The droplet diameter and spatial distribution indicates that SMD decreases in the spray intermediate area with the increase in the aerodynamic force action.

(4) The diameter of droplets ranges from 5 μm to 1000 μm under Test 1 operating condition, indicating as a unimodal-normal distribution. With the enhance of aerodynamic force, the droplet diameter converts to bimodal-normal distribution (Test 2) with large droplets individually, then returned to unimodal-normal distribution includes all small droplets (Test 3).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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